

# Lecture 4: Optimal Prediction and Holt-Winters Forecasting

Ignasi Cos

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## 1 Introduction

Assume we have  $n$  variables  $X_1, \dots, X_n$ . Denote the realized data by  $x_1, \dots, x_n$ .

Assume we wish to predict the value of a non-accounted (observed) random variable  $Y$ . In the context of time-series this may mean, for example, to predict  $X_{n+1}, X_{n+p}$  or  $f(X_{n+p})$  for a certain function  $f$ . The question is, given  $x_1, \dots, x_n$ , what can be inferred about  $Y$ ?

In the second order framework, the natural answer is to search for a function

$$f(X_1, \dots, X_n)$$

such that

$$\mathbb{E}[(Y - f(X_1, \dots, X_n))^2]$$

is minimal.

It is well-known, see for example [1] (Section 2.7), that the solution to this problem is the conditional expectation, i.e.,

$$f(X_1, \dots, X_n) = \mathbb{E}[Y|X_1, \dots, X_n]$$

which is the so-called Optimal Predictor of  $Y$  given  $X_1, \dots, X_n$ .

However, to compute the conditional expectation is often non-obvious. Thus, the search for a predictor of  $Y$  is often restricted to the class of affine functions

$$f(X_1, \dots, X_n) = b_0 + b_1 X_1 + \dots + b_n X_n$$

and the problem reduced to the search of the  $b_i$  coefficients such that

$$\mathbb{E}[(Y - b_0 - b_1 X_1 - \dots - b_n X_n)^2]$$

is minimal. The solution is the so-called optimal linear predictor of  $Y$  in terms of  $X_1, \dots, X_n$ . It may be expressed as

$$P(Y|X_1, \dots, X_n) = \hat{b}_0 + \hat{b}_1 X_1 + \dots + \hat{b}_n X_n$$

Naturally, the expressions

$$\mathbb{E}[(Y - E(Y|X_1, \dots, X_n))^2]$$

and

$$\mathbb{E}[(Y - P(Y|X_1, \dots, X_n))^2]$$

are the matching prediction errors.

The underlying idea behind the notion of prediction under a second order framework is described next. Consider a Hilbert space (or Euclidean space)  $H$ , with an associated scalar product and orthogonality definition. Consider a Hilbert subspace  $G \subset H$ . It is well-known that the best prediction of  $h \in H$  by elements of  $G$  is the element  $g := P_G h \in G$ . In other words, the orthogonal projection of  $h$  onto  $G$ . That projection exists and is unique.

## 2 Example

Throw two dices, one after another. Assume we are interested in the following variables:

$X$  : result of the first dice

and

$Y$  : squared sum of the two outcomes

The question we are interested in is the following: after observing the first result, the result of  $X$ , what is the best prediction of  $Y$ ?

Let  $X$  and  $X'$  the results of the two dices and  $Y = (X + X')^2$  the squared sum of the two results.

Compute first the optimal predictor:

$$\begin{aligned}\mathbb{E}[Y|X = x] &= E[(X + X')^2|X = x] = E[(x + X')^2|X = x] = \\ &= x^2 + E[X'^2|X = x] + 2xE[X'|X = x] = \\ &= x^2 + E[X'^2] + 2xE[X']\end{aligned}$$

Since we have  $\mathbb{E}(X') = 3.5$  and  $\mathbb{E}((X')^2) = 15.167$ ,

$$E[Y|X = x] = x^2 + 7x + 15.167$$

and, as a RV, we have:

$$E[Y|X] = X^2 + 7X + 15.167$$

Compute now the optimal linear estimator. We have

$$P(Y|X) = b_0 + b_1X$$

and we search for  $\hat{b}_0$  and  $\hat{b}_1$  such that

$$\mathbb{E}[Y - \hat{b}_0 + \hat{b}_1X]^2]$$

is minimal.

The traditional method to solve this problem consists of first considering a function  $\varphi$  defined on  $\Re^2$  such that

$$\begin{aligned}\varphi(b_0, b_1) &= \mathbb{E}[(Y - b_0 - b_1X)^2] = \\ &= \mathbb{E}[(Y - b_0)^2] + \mathbb{E}[X^2] - 2(b_1)\mathbb{E}[(Y - b_0)X] = \\ &= \mathbb{E}(Y^2) + (b_0)^2 - 2(b_0)\mathbb{E}[Y] + (b_1)^2\mathbb{E}(X^2) - 2b_1\mathbb{E}(YX) + 2b_0b_1\mathbb{E}(X)\end{aligned}$$

to compute its partial derivatives

$$(\partial_{b_0}\varphi)(b_0, b_1) = 2b_0 - 2\mathbb{E}(Y) + 2b_1\mathbb{E}(X)$$

$$(\partial_{b_1} \varphi)(b_0, b_1) = 2b_1 \mathbb{E}(X^2) - 2\mathbb{E}(XY) + 2b_0 \mathbb{E}(X)$$

and equate them to 0 in to find the optimal  $(\hat{b}_0, \hat{b}_1) \in \mathbb{R}^2$  that minimizes  $\varphi$ . Compute the matrix of second derivatives and check it is positive definite (determinant is positive). Specifically, it equals  $4V(X)$

We have

$$\begin{aligned} 2b_0 + 2\mathbb{E}(X)b_1 &= 2\mathbb{E}(Y) \\ 2\mathbb{E}(X)b_0 + 2\mathbb{E}(X^2)b_1 &= 2\mathbb{E}(XY), \end{aligned}$$

that is

$$\begin{bmatrix} 1 & \mathbb{E}X \\ \mathbb{E}X & \mathbb{E}(X^2) \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} = \begin{bmatrix} \mathbb{E}X \\ \mathbb{E}(XY) \end{bmatrix}$$

Writing  $\mu_X = \mathbb{E}X$ ,  $\mu_Y = \mathbb{E}Y$ ,  $\sigma_X^2 + \mu_X^2 = \mathbb{E}(X^2)$  and  $\mu_{XY} + \mu_X\mu_Y = \mathbb{E}(XY)$ , we have

$$\begin{bmatrix} \hat{b}_0 \\ \hat{b}_1 \end{bmatrix} = \begin{bmatrix} 1 & \mu_X \\ \mu_X & \mu_X^2 + \sigma_X^2 \end{bmatrix}^{-1} \begin{bmatrix} \mu_Y \\ \sigma_{XY} + \mu_X\mu_Y \end{bmatrix}$$

The inverse matrix is

$$\frac{1}{\sigma_X^2} \begin{bmatrix} \mu_X^2 & -\mu_X \\ -\mu_X & 1 \end{bmatrix}$$

Therefore,

$$\hat{b}_0 = (1 + \frac{\mu_X^2}{\sigma_X^2})\mu_Y - \frac{\mu_X}{\sigma_X^2}(\sigma_{XY} + \mu_X\mu_Y) = \mu_Y - \frac{\sigma_{XY}}{\sigma_X^2}\mu_X$$

Note that

$$\hat{b}_0 = \mu_Y - \hat{b}_1\mu_X$$

In our case, we have  $\mu_X = 3.5$ ,  $\sigma_X^2 = 15.167 - 3.5^2 = 2.917$ ,

$$\begin{aligned} \mathbb{E}Y &= \mathbb{E}[(X + X')^2] = \mathbb{E}[X^2] + \mathbb{E}[(X')^2] + 2\mathbb{E}[X]\mathbb{E}[X'] = \\ &= 2(\mu_X^2 + \sigma_X^2) + 2\sigma_X^2 = \\ &= 4\mu_X^2 + 2\sigma_X^2 = \\ &= 43.5^2 + 22.917 = \\ &= 54.834 \end{aligned}$$

and

$$\begin{aligned} \sigma_{XY} &= \mathbb{E}(XY) - \mu_X\mu_Y = \\ &= \mathbb{E}[X(X + X')^2] - \mu_X\mu_Y = \\ &= \mathbb{E}(X^3) + \mathbb{E}[X(X')^2] + 2\mathbb{E}[X^2X'] - \mu_X\mu_Y = \\ &= \mathbb{E}(X^3) + \mu_X\mathbb{E}[(X')^2] + 2\mu_X\mathbb{E}[X^2] - \mu_X\mu_Y = \\ &= \mathbb{E}(X^3) + 3\mu_X(\sigma_X^2 + \mu_X^2) - \mu_X\mu_Y = \\ &= 73.5 + 33.5(2.917 + 3.5^2) - 3.554.834 = \\ &= 40.835 \end{aligned}$$

Finally,

$$\widehat{b}_0 = 13.999\widehat{b}_1 = 5.837$$

and therefore

$$P(Y|X) = 13.999X + 5.837$$

Prediction errors are:

$$\mathbb{V}(Y - E(Y|X)) = \mathbb{E}[(Y - X^2 - 7X - 15.167)^2] = 611.908$$

$$\mathbb{V}(Y - P(Y|X)) = \mathbb{E}[(Y - 13.999X - 5.837)^2] = 11152.652$$

Note that these errors may be expressed as polynomials of the four first moments of X, namely  $\mathbb{E}(X) = 3.5$ ,  $\mathbb{E}(X^2) = 15.167$ ,  $\mathbb{E}(X^3) = 73.5$ ,  $\mathbb{E}(X^4) = 379.167$ . For example, if  $X = 3$ , the variable  $Y$ , conditioned to  $X = 3$  assumes values within the set  $\{16, 25, 36, 49, 64, 81\}$  with equal probability. Thus, the expected value is 46.167. The optimal prediction yields exactly the same value, and the linear best approximation 47.834. If, by contrast,  $X=1$ , this yields the set  $\{4, 9, 16, 25, 36, 64\}$ , with expectation 23.667, equal to the optimal prediction. The best linear prediction is 19.836. It is an exercise of interest to compute each case and to draw the prediction for both functions.

### 3 The Optimal Linear Predictor

In the general case we have the following expression for the optimal linear predictor:

$$P(Y|X) = \mu_Y + \frac{\sigma_{XY}}{\sigma_X^2}(X - \mu_X)$$

Note that to compute  $\widehat{b}_0$  and  $\widehat{b}_1$  consists in searching  $Z$  such that  $Z = b_0 + b_1X$  and  $E[(X - Z)^2]$  is minimal. It is known that the solution of this problem is the orthogonal projection of  $Y$  over the linear subspace of  $L^2(\Omega)$  generated by 1 and  $X$ .

Thus, if  $Y$  has an orthogonal 1 and  $X$ , it satisfies:

$$\mathbb{E}[(Y - Z)1] = 0$$

and

$$\mathbb{E}[(Y - Z)X] = 0$$

and so,

$$\mathbb{E}[(Y - b_0 - b_1X)] = 0$$

$$\mathbb{E}[(Y - b_0 - b_1X)X] = 0$$

and finally

$$b_0 + \mathbb{E}[X]b_1 = \mathbb{E}[Y]$$

$$\mathbb{E}[X]b_0 + \mathbb{E}[X^2]b_1 = \mathbb{E}[XY]$$

Note that these are the same equations obtained in the previous section.

## 4 Exponential smoothing and Holt-Winters forecasting

A particular case of filtering typically used to predict is the so-called exponential smoothing, a method related with the well-known Holt-Winters prediction algorithm. Imagine we have a time-series  $x_1, x_2, \dots, x_n$  with no systematic trend or seasonal effects and with no systematic trend or seasonal effects and we want to predict  $x_{n+h}$  with  $h \geq 1$ . For example, the sale time-series of a well-established product in a stable market.

The method of exponential smoothing proposes to estimate a trend process  $m$  recursively, defining  $m_1 := x_1$  and

$$m_n = ax_n + (1 - a)m_{n-1}, n \geq 2,$$

with  $a \in (0, 1)$ .

Note that this implies of one hand,

$$m_n = ax_n + a(x_n - m_{n-1}), n \geq 2$$

and on other hand,

$$m_n = \sum_{j=0}^{n-2} a(1-a)^j x_{n-j} + (1-a)^{n-1} x_1.$$

Observe that the exponential smoothing is a weighted average of the previous data, with decreasing weights in the past.

The parameter  $a$  can be chosen or estimated. Note that if  $a$  is close to 1, the trend is almost equal to the series. By contrast, the closer  $a$  to 0, the more the estimated trend  $m$  changes very slowly from  $x_1$ . The closer  $a$  to 0 the smoother is the estimated trend  $m$ . A typical value is  $a = 0.2$ .

By default, the best estimation of  $x_n$  is  $m_{n-1}$ . This fact suggests that parameter  $a$  can be estimated by minimizing the error

$$E_n(a) := \sum_{j=2}^n (x_j - P_{j-1}x_j)^2 = \sum_{j=2}^n (x_j - m_{j-1})^2$$

In general  $P_n x_{n+h} = m_n$  for any  $h \geq 1$ .

Holt — see reference, generalised the method to include time-series with trends. The formulae are:

$$m_n = ax_n + (1 - a)(m_{n-1} + t_{n-1})$$

and

$$t_n = b(m_n - m_{n-1}) + (1 - b)t_{n-1}$$

for  $n \geq 2$  with  $m_1 = x_1$  and  $t_1 = 0$ .

In this case, the prediction is given by  $P_n x_{n+h} = m_n + t_n$  for any  $h \geq 1$ .

The error, as before, is given by:

$$E_n(a, b) = \sum_{j=2}^n (x_j - P_{j-1}x_j)^2 = \sum_{j=2}^n (x_j - m_{j-1} - t_{j-1})^2$$

Finally, Winters, in 1960, made an extension is given by:

$$P_n x_{n+h} = m_n + ht_n + s_{n+h-p}$$

and the error by:

$$E_n(a, b, c) = \sum_{j=2}^n (x_j - P_{j-1} x_j)^2 = \sum_{j=2}^n (x_j - m_{j-1} - t_{j-1} - s_{j-p})^2$$

## References

- [1] P. J. Brockwell and R. A. Davis (1991): *Time Series: Theory and Methods*. Springer.